

Healthcare IT Fundamentals – Section 2,

Lecture 2

Clinical Workflows & Data Integration (Simple Notes)

Big Picture

In the previous lecture, we learned:

- Components of an EHR
- CPOE
- Patient Portals
- Clinical Decision Support (CDS)
- PACS
- Billing modules

Now we learn:

How data actually flows through the hospital during a patient visit.

Think of this lecture as:

"The life journey of patient data inside an EHR."

1. Patient Journey in an EHR

A patient's healthcare journey starts before they even enter the hospital.

Step 1: Pre-Visit

Patient uses:

- Website
- Mobile app
- Patient portal

to:

- Schedule appointment
 - Fill forms
 - Update medical history
-

Example

Patient enters:

```
Reason for Visit:  
Chest pain
```

This information already appears in EHR before arrival.

Step 2: Registration

Front desk verifies:

- Name
- Date of Birth
- Address
- Insurance

System creates an encounter record.

Why Important?

Wrong registration can lead to:

✗ Wrong patient record

✗ Billing issues

✗ Patient safety risks

Step 3: Intake

Nurse records:

- Blood pressure
- Temperature
- Weight
- Symptoms

Example

BP: 140/90

Temp: 101°F

Weight: 75kg

Doctor can immediately see this data.

Step 4: Clinical Encounter

Doctor enters:

- Diagnoses
- Notes
- Treatment plans

into EHR.

Step 5: Ordering

Doctor uses CPOE to order:

- Blood tests
 - MRI
 - X-ray
 - Medications
-

Step 6: Results

Lab and radiology systems return results automatically.

Step 7: Discharge

System generates:

- Visit summary
- Instructions
- Prescriptions

and sends them to patient portal.

Complete Workflow

```
graph TD; A[Pre-Visit] --> B[Registration]; B --> C[Nurse Intake]; C --> D[Doctor Visit]; D --> E[Orders]; E --> F[Results]; F --> G[Treatment]; G --> H[Discharge]; H --> I[Patient Portal];
```

Key Point

An EHR is NOT just a database.

It is a:

Workflow Engine

that guides the patient through every stage.

2. Structured vs Unstructured Data

One of the most important concepts in Healthcare IT.

Structured Data

Data stored in predefined fields.

Examples:

- Dropdowns
 - Checkboxes
 - Radio buttons
 - Numeric fields
-

Example

Gender:

☒ Male

☐ Female

Or

Blood Pressure:

140/90

Benefits

Computers can easily:

- Search
 - Filter
 - Report
 - Analyze
-

Example

Find all diabetic patients:

```
SELECT *  
FROM patients  
WHERE diagnosis='diabetes'
```

Easy because data is structured.

Unstructured Data

Free-text notes.

Written by humans.

Example

```
Patient reports occasional chest discomfort  
during exercise and mild dizziness.
```

Humans understand it easily.

Computers struggle.

Why Unstructured Data is Hard

A computer cannot easily answer:

How many patients have chest pain?

when information is buried inside paragraphs.

Requires:

NLP (Natural Language Processing)

to analyze.

Comparison

Structured Data	Unstructured Data
Checkboxes	Paragraphs
Easy to Analyze	Hard to Analyze
Machine Friendly	Human Friendly
Reporting Ready	Requires NLP

Why Healthcare Loves Structured Data

Structured data enables:

- ✔ Quality reporting
- ✔ Analytics
- ✔ Clinical Decision Support
- ✔ Population health studies
- ✔ Regulatory compliance

3. Real-Time Documentation

Traditionally doctors documented later.

Old Method

Doctor sees patient.

Hours later:

`Writes notes from memory`

Problems:

- Missing details
 - Mistakes
 - Delays
-

Modern Method

Document immediately.

At point of care.

Example

Doctor enters diagnosis while patient is still present.

Benefits

Better Accuracy

Fresh information.

Less forgetting.

Better Communication

Nurses immediately see doctor's plan.

Better Safety

Entire care team sees latest information.

Faster Billing

Claims can be generated sooner.

Pajama Time

Interesting healthcare term.

Meaning

Doctors finish documentation at home after work.

Usually at night.

Hence:

Pajama Time

Goal of Modern EHR

Reduce pajama time through:

- Mobile devices

- Tablets
 - Bedside documentation
-

4. Importance of Demographics & Insurance Data

Many people think clinical data is most important.

Actually:

Administrative data is equally important.

Demographics

Includes:

- Name
 - DOB
 - Address
 - Contact info
-

Why Important?

Correct patient identification.

Example

Two patients:

John Smith

John Smith

Wrong DOB entered.

Records get mixed.

Potentially dangerous.

Master Patient Index (MPI)

Master system used to uniquely identify patients.

Think:

User Table

for the entire healthcare organization.

Purpose

Avoid duplicate records.

Maintain one patient identity.

Insurance Verification

Verifies:

- Coverage
 - Eligibility
 - Benefits
-

Example

Wrong insurance number entered.

Result:

- ✗ Claim denied
- ✗ Delayed payment
- ✗ Extra administrative work

5. Social Determinants of Health (SDOH)

Healthcare now captures social factors too.

Examples:

- Income level
 - Housing status
 - Transportation access
 - Food security
-

Why?

These factors strongly affect health outcomes.

Example

Patient misses appointments.

Reason:

No transportation.

Not because they ignore medical advice.

6. Interdepartmental Data Sharing

One of the biggest advantages of EHR.

Before Integration

Departments operated separately.

Doctor

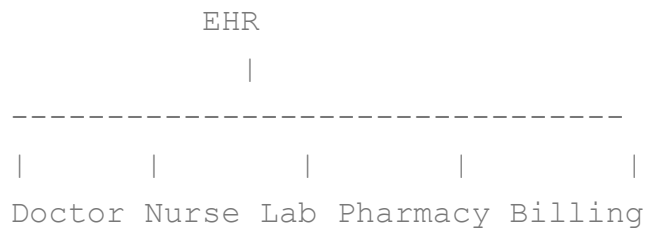
|

Pharmacy
|
Radiology
|
Billing

Poor communication.

After Integration

Everyone uses the same system.



Single source of truth.

Example: Pharmacy

Doctor prescribes medication.

Pharmacy instantly sees:

- Diagnosis
 - Medication order
 - Allergies
-

Example: Radiology

Technician sees:

- Clinical history
- Prior scans

- Relevant lab values
-

Example: Billing

Billing team sees:

- Diagnoses
- Procedures
- Documentation

No need to call physician repeatedly.

Benefits

- ✓ Faster care
 - ✓ Better communication
 - ✓ Fewer mistakes
 - ✓ Less duplication
-

7. Quality Reporting

Healthcare organizations must report quality metrics.

Examples:

- Readmission rates
 - Infection rates
 - Mammogram compliance
 - Diabetes management
-

These metrics are automatically extracted from EHR.

Example

Patient:

Diabetic

Provider performs:

Foot Examination

Correct Documentation

Checkbox selected:

☒ Foot Exam Completed

System counts it.

Incorrect Documentation

Doctor writes:

Performed diabetic foot exam.

inside free-text note.

System may miss it.

Report shows:

Foot Exam NOT Performed

even though it happened.

Important Lesson

Good care + Bad documentation

≠ Good quality score

Feedback Loop

Healthcare organizations constantly work to improve documentation quality.

Goal:

Actual Care
=
Recorded Care

Why?

Because:

- Compliance depends on it
 - Reporting depends on it
 - Reimbursement depends on it
-

Quick Revision Sheet

Patient Journey

Pre-Visit → Registration → Intake → Provider → Orders → Results → Discharge

Structured Data

Checkboxes, dropdowns, fields.

Easy for computers.

Unstructured Data

Free-text notes.

Easy for humans.

Hard for computers.

Real-Time Documentation

Document during care delivery.

Pajama Time

Doctors doing documentation at home after work.

MPI

Master Patient Index.

Unique patient identity system.

Insurance Verification

Validates coverage and eligibility.

SDOH

Social Determinants of Health.

Factors like housing, food, income, transportation.

Data Sharing

Enables collaboration across departments.

Quality Reporting

Uses structured EHR data to measure healthcare performance.

For a Ruby on Rails Developer

Think of the workflow like this:

Software Concept	Healthcare Equivalent
User Signup	Registration
User ID	MPI
Form Fields	Structured Data
Text Area	Unstructured Data
Event Pipeline	Clinical Workflow
Shared Database	Integrated EHR
Analytics Dashboard	Quality Reporting
Background Jobs	Lab Result Processing
Audit Data	Compliance Reporting

One-Line Takeaway

The quality of healthcare depends heavily on the quality of data entered into the EHR. Structured, accurate, real-time documentation powers patient safety, reporting, interoperability, and reimbursement. 🚀